Identification of stakeholder-specific Systems Engineering competencies for industry

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***Abstract*—This paper presents a concept to identify Systems Engineering (SE) competencies for different stakeholder groups within the company. An overview of the current state of research and the challenges of implementing Systems Engineering in industry is given. The qualification of employees towards the topic of SE has a very decisive role for the company-wide stabilization of SE. In order to gain the acceptance for SE among the employees, the individual needs must be addressed. Different competencies are required depending on the area of responsibility.**

Keywords—systems engineering, competencies, stakeholder, qualification, skills, abilities, transformation

I. Introduction

Product complexity is increasing due to various drivers, such as digitalization. Products with new technologies are often referred to as **Advanced Systems** [1]. These systems are e.g., autonomous systems which can solve complex tasks independently and manage unexpected situations. Furthermore, they are interactive socio-technical systems with the ability to interact with other systems (System-of-Systems). That means that they are user-friendly and consider the interaction between human and machine. Another trend are new product-service systems. Tailored solutions will provide products and data-driven services that include data collection, processing, and analysis [2].

Due to increasing product complexity, the associated product development is also becoming increasingly complex. **Systems Engineering (SE)** is a proven approach to meet these challenges [3]. "Systems Engineering is a transdisciplinary and integrative approach to enable the successful realization, use, and retirement of engineered systems, using systems principles and concepts, and scientific, technological, and management methods" [4].

Various studies have shown that **introducing Systems Engineering company-wide** is a major challenge for enterprises [5], [6]. Often, companies are unsure how to start with the implementation and optimization process. There is a lack of a strategic approach to the company-wide introduction and steadiness of SE. Many questions arise within this context that need to be addressed. For this purpose, **a holistic approach** is recommended [3], which takes into account the aspects of **"human, technology and organization"**. Based on the studies and analysis of the challenges in the introduction and steadification of SE, the need for action was identified and the research project SE4OWL was initiated. The need for action is described in detail in [3] and provides the starting point for the current paper. In the research project, following research questions are considered: How can methods and tools be adapted to be stakeholder-oriented? How can companies create ITS compatible engineering structures? How can individual and organizational acceptance be achieved in companies?

This paper provides insight into the initial research findings within the research project, specifically considering the aspects of individuals' acceptance. The aim is to integrate and disseminate Systems Engineering to stakeholders using new learning methods. The needs of the stakeholders with regard to the depth and breadth of the content must be taken into account. To this end, it is necessary to select formats that are suitable for stakeholders, to design them, and to anchor them in the companies' qualification programs. Scaling and a company-neutral design should ensure transferability to other companies. To be able to develop these qualification plans tailored to the needs of the stakeholders, the first step is to ask about individual competencies: **Which Systems Engineering competencies are required by which employees?**

Furthermore, companies are adopting agile approaches to be able to react flexibly to newly gained knowledge regarding customer and user requirements [2]. Additionally, the competencies and roles required for **agile development** will be considered in this paper. The consideration of agile competencies together with SE competencies was expressed by the industry partners of the research project as an additional challenge.

In Germany the empowerment of users is currently carried out via an increasing number of **training providers**. Basics as well as methods and tools are taught in combination with best practices in a general matter. Developers, especially system engineers, are usually defined as the main target group among all stakeholders. However, in order to establish SE in the company, content and concepts must be developed specifically for a company and all of its stakeholders on an individual basis.

Qualification offerings are becoming more individualized and should be available at all times to ensure the link between competence achievement and application of trained skills. Transfer of training content to daily work is not ensured [2]. Learning transfer is the basis of successful qualification but actually takes place in only 10 % to 30 % of training measures [7]. Continuous adaptation of competence profiles with associated qualification measures is essential. The need for competencies develops on a highly individual basis. Digitalization and new ways of working support the stated need of individual and readily available qualification offerings

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[8]. To ensure individual and stakeholder-specific qualification plans it is essential to understand which competencies are needed by which stakeholder and which competencies individuals already have. This paper will focus on the identification of needed competencies for different stakeholders.

II. BACKROUND AND THEORY

To define stakeholder-specific Systems Engineering competencies, we have analyzed current approaches and existing models. First, we compared and analyzed existing competency frameworks and models of Systems Engineering. Furthermore, we have taken into account available Systems Engineering role models.

A. Competencies

The INCOSE presents the **INCOSE Systems Engineering Competency Framework** [9], which is a framework that offers a set of competencies in the form of knowledge, skills, abilities, and behaviors important for the effective usage of Systems Engineering. The purpose of the framework is to support the identification and definition of competencies. Five categories of competencies are defined: core, professional, management, technical, and integrating competencies. In each category, between four and nine competencies are defined as shown in Figure 1.

CORE COMPETENCIES	PROFESSIONAL COMPETENCIES	MANAGEMENT COMPETENCIES	TECHNICAL COMPETENCIES	INTEGRATING COMPETENCIES
Systems Thinking	Communications	Planning	Requirements Definition	Project Management
Lifecycles	Ethics and Professionalism	Monitoring and Control	System Architecting	Finance
Capability Engineering	Technical Leadership	Decision Management	Design for…	Logistics
General Engineering	Negotiation	Concurrent Engineering	Integration	Quality
Critical Thinking	Team Dynamics	Business and Enterprise Integration	Interfaces	
Systems Modeling and Analysis	Facilitation	Acquisition and Supply	Verification	
	Emotional Intelligence	Information Management	Validation	
	Coaching and Mentoring	Configuration management	Transition	
		Risk and Opportunity Management	Operation and Support	

Fig. 1. Competencies according to INCOSE Competency Framework

Each of those competencies is divided into five levels of increasing proficiency starting with "Awareness" over "Supervised Practitioner", "Practitioner", "Lead Practitioner" up to "Expert". Moreover, the framework provides usage scenarios based on the typical user of the framework and a rough tailoring concept for the framework. The defined competencies should be applicable to all kinds of companies and organizations. Lastly, a link between the competencies and the SE processes of the ISO 15288 has been made, expect for the "Professional Competencies".

Valuation: The framework of the INCOSE is on a detailed level and covers all competencies according to SE. However, other important competencies for the future development like agile aspects are missing. Furthermore, the matching of competencies to stakeholders is missing. There is no indication how those competencies shall be identified for each stakeholder. The called "Professional Competencies" are not assigned to any processes. Other important processes, such as support and qualification, have not been taken into account.

LESLIE defined an **Engineering Competency Model** which identifies the knowledge, skills, and abilities needed to successfully perform the work in fields of engineering. Figure 2 shows the structure of the model [10]. The model is divided into six tiers and the pyramid shape represents the increasing specialization and specificity of each tier. The tiers one to three represent the soft skills and work readiness competencies. The competencies in tier four and five include the industry-specific competencies which are necessary for career lattices within a specific industry.

The first tier "Personal Effectiveness Competencies" contributes to the personal attributes, like emotional intelligence or open communication. The second tier "Academic Competencies" includes those skills which have been learned in school. They include cognitive functions and thinking styles. Tier three "Workplace Competencies" includes abilities like self-organization and teamwork. Tier four "Industry-Wide Competencies" covers the knowledge, skills, and abilities from which workers within an industry can benefit, independent of the sector they are working in. Tier five "Industry-Sector Functional Areas" represents a sub-set of industry-wide competencies that are specific to an industry sector. The last tier "Management & Occupation-Specific Competencies" represents further competencies that are necessary for managing organizations and projects like staffing or strategic planning.

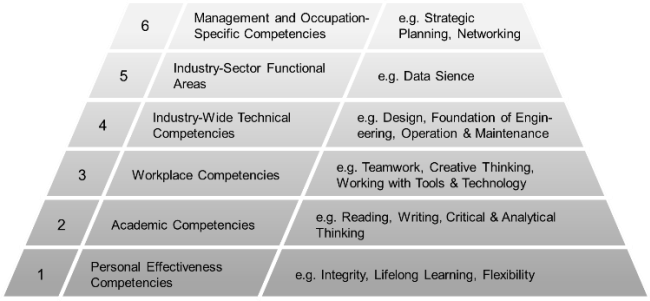

Fig. 2. Engineering Competency Model according to LESLIE

Valuation: The Engineering Competency Model of LESLIE is a holistic and well-structured model that considers the different levels of competencies in engineering. Systems Engineering has been considered in tier four. However, the Systems Engineering competencies are only considered on a high level, which is not sufficient to define SE competencies. Moreover, agile competencies are not considered at all. Finally, the model is not considering roles nor the allocation of competencies to roles.

B. Roles

Based on publications within the Systems Engineering Journal of INCOSE, SHEARD identified more than 60 role descriptions and grouped them into twelve categories, which are briefly described in Table I. Each role has been given a name and the responsibilities in the context of Systems Engineering have been defined. An example is the "System Analyst" who is responsible for performing simulations and ensuring that the system will meet the requirements (cf. Table I). The results were checked for plausibility and completeness by comparing them to the proceedings of the INCOSE symposium as well as job advertisements. The identified roles should be applicable to typical aerospace or department of defense environments as well as commercial companies [11].

TABLE I. 12 SYSTEMS ENGINEERING ROLES [5]

Role Name	Responsibility
Requirements Owner (RO)	• Derive requirements from customer needs • Create functional architecture

	• Analyze change impact • Specify sub systems
System Designer (SD)	• Selection of main elements of the system • Specification of the lower sub system layer
System Analyst (SA)	• Perform simulations • Ensure that the system will meet the requirements
V & V (VV)	• Plan and implement the verification program
Logistics & Operations (LO)	• Represents the later life cycle phases (service, logistics, operations, disposal) • Responsible for maintenance or upcoming issues during operations
Glue (G)	• Integration of the sub systems • Responsible for all internal interfaces • Prevent that sub systems interfere with each other in a disturbing way
Customer Interface (CI)	• Communication with the customer • Representing the customer within the project
Technical Manager (TM)	• Planning and allocation of resources • Control of cost
Information Manager (IM)	• Planning and enabling of configuration management and data management
Process Engineer (PE)	• Documents and improves the SE process • Ensures the fulfilment of the process
Coordinator (CO)	• Mediates in case of conflicts • Moderates
Others	• Represents the rest of the roles, based on job advertisements

Valuation: The role descriptions of SHEARD are on a detailed level and cover relevant SE tasks within a development project. The roles show the wide range of responsibilities that come with SE within a project. However, roles outside a project context, for example support of IT infrastructure, are missing. The only exception is the "Process Engineer". Also, aspects of agile development are not covered explicitly, which is understandable, as agile is a younger trend. Competencies are not considered in this model.

BRETZ has developed an adapted role model based on SHEARD [11] and FRIEDENTHAL [12]. There is often debate about whether new roles in the organization are needed for the tasks or whether existing roles should take on new tasks. [13] BRETZ distinguishes between "**project roles**" and "**supporting roles**" (cf. Figure 3). Project roles are the „classic roles" of the system engineer. Supporting roles describe tasks that enable the project work by providing processes, tools, services, and support.

Validation: BRETZ makes a distinction between supporting roles and project roles. The distinction offers approaches with regard to differentiation of SE competencies and the associated responsibilities. However, a requirement of the research project is that existing roles shall be considered within the role model. Companies want to avoid the establishment of new roles when introducing SE. Since the roles defined by BRETZ are unknown to those companies, a focus has to be put on established roles. The defined roles of BRETZ do not consider further engineering related roles, such as a product or production manager. Moreover, agile roles like a product owner are not considered either. Finally, there is no link to necessary competencies.

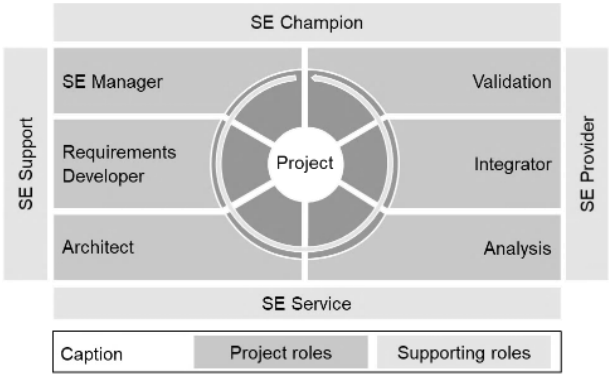

Fig. 3. Role model according to [13]

III. CONCEPT

The analyzed state of the art shows that the current approaches are not sufficient to generate a usable role set with allocated competencies which can be tailored for industry demands. Therefore, we will introduce role clusters which consider all relevant stakeholders of Systems Engineering and their responsible tasks (cf. III A). Based on this, we have presented a matrix where for each task the necessary competencies are allocated. Furthermore, a matrix has been derived which defines which role needs which competencies (cf. III B). Finally, the validation results will be presented, and further steps will be defined.

A. Consideration of the stakeholders

As part of the research project, the existing role descriptions (cf. II "Valuation") were reflected. Since a company-wide introduction of SE can only succeed if there is **acceptance and understanding by all employees in the company**, the first step was to analyze which stakeholders operate in the company and which tasks have to be performed by them. To this purpose, the existing job titles were collected from three industrial companies.

There is a trend among companies to give the individual roles new modern job titles. ULMER says that the reason can be found in today's world of rapid change. Because job descriptions are adjusted to reflect a situational picture of the job, they are difficult to keep current [14]. It was noted in the project consortium that many job titles are based on a common understanding. ULMER further explains that it is necessary to concentrate on the question: What is the job supposed to achieve, what role is assigned to it? From this, the tasks often arise by themselves as a logical consequence. For this reason, business and science are using a new term: the role model.

In the research project, role clusters were defined in order to group together roles with similar job descriptions. **14 role clusters** (cf. Table II) could be identified and validated from the industry companies with the assignment of the roles into the clusters. An example is the cluster "customer representative". This cluster forms the interface between the customer and the company. The roles in this cluster form the voice for all customer-relevant information required for the project. Typical tasks performed by this role cluster are: Uncovering the stakeholder and its requirements, making decisions about critical or desired system functionalities, prioritizing, and review, maintain and manage requirements. The following role names might be in use in the company for

the cluster and perform the tasks: product manager, requirements owner, product owner or strategic marketing. In special machinery, the technical sales department often performs these tasks. To illustrate a representative employee of the cluster, in the following example, the requirements owner is used. The requirements owner has the task of identifying and understanding stakeholder requirements and translating them into technical requirements. To do this, he or she is in constant communication with internal and external stakeholders. The requirements manager is responsible for communicating any requirements that may change during the course of the project and for tracking their impact on subsystems and the overall system.

TABLE II. ROLE CLUSTERS

Role Cluster Name	Description
Customer	represents the party that orders or uses a service (= development order). The customer has influence on the design/technical execution of the system.
Customer representative	forms the interface between the customer and the company. The roles in this cluster form the voice for all customer-relevant information required for the project.
Project manager	is responsible for the planning and coordination on the project side. The roles assume responsibility for achieving the project goals and monitoring the resources (time, costs, personnel) within a time-limited framework and also have a moderating role in conflicts and disputes.
System developer	has the overview from requirements to the decomposition of the system to the interfaces and the associated system elements (external to the system environment and internal between the elements). The system developer is responsible for integration planning and consults with the appropriate subject matter experts.
Specialist developer	includes the various specialist areas, e.g., software, hardware, etc. They develop new technologies or realize the product/system on the basis of specifications from the system developer cluster.
Production planner / coordinator	takes on the preparation of the product realization and the transfer to the customer.
Production employee	comprises the processes that are to be assigned to the implementation, assembly, and manufacture of the product through to goods issue and shipping. The individual system components are integrated into the overall system and verified with regard to their functionality.
Quality engineer / manager	ensures that the company's quality standards are maintained in order to keep customer satisfaction high and ensure long-term competitiveness in the market. Close cooperation with the V&V operator, e.g. for the analysis of customer complaints and identification of the cause.
Verification and validation (V&V) operator	covers the topics of system verification & validation. The involvement of this role cluster in the early phases of system development can ensure that the system is verifiable and validatable.
Service technician	deals with all service-related tasks at the customer's site, i.e. installation, commissioning, professional training of users, as well as classic service tasks such as maintenance and repairs, or the area of after-sales.
Process and policy manager	is divided into a strategic and an operational level: On a strategic level, the process owner serves to develop internal guidelines in the development and creation or revision of process flows. On an operational level, the policy owner controls compliance with policies, laws and framework conditions that must be taken into account and fulfilled.
Internal support	represents the advisory and supporting side during the development process within the project. A distinction is made between: - IT support: IT support provides and maintains the necessary IT infrastructure. - Qualification support: On the one hand, this provides support in the area of methods and, on the other hand, the qualification of the employees is individually ensured by means of specialized training. This can be done by the HR department, which also supports the project by acquiring suitable employees. - Systems Engineering (SE) support: The SE support offers separate support with regard to SE methods and handling of SE tools. This offers assistance in order to impart the necessary knowledge in the SE procedure.
Innovation management	focuses on the commercially successful implementation of products or services, but also new business models or processes.
Management	forms the group of decision-makers and is represented by the management or department management. The cluster keeps an eye on the company's goals, visions, and values. Since the opinion of the cluster is crucial for project progress, management is an important stakeholder in every respect.

Likewise, it was examined how the term "**role**" should be considered in the context of the research project and which aspects are relevant for the introduction of SE.. A role has "**tasks**" to be performed in a field of activity. A distinction is made between **tasks for which the role owner is primarily responsible and tasks that are supported by the role in a process.** A role holder also has its own "personality" structures. These **personality structures** must be taken into account in order to motivate the role and thus generate acceptance for the topic of SE. In addition, knowledge of the personality structures that a role should have for a position can help in the hiring process.

Based on the definition of the aspects, it is clear that the individual roles in companies require **different depth and breadth of knowledge (= competencies)** around Systems Engineering. With the formation of the role clusters, different stakeholder-appropriate measures can be derived for the clusters in terms of **qualification and communication / motivation concepts**.

In order to describe each role cluster in detail, the tasks of the individual role clusters were described in the further course of the project. For this purpose, the tasks of the existing ISO 15288 [15] were assigned to the role. The ISO 15288 is an important standard and provides a good overview about SE processes. The existing processes of the ISO 15288 were also complemented by further tasks. For this purpose, in cooperation with the industrial companies, the areas "support,

moderation, qualification, and training" were added. The assignment of role clusters to processes (cf. Figure 4) was carried out with the classification into "is responsible" (= 2), "supports the process step" (= 1) or is not involved (= 0 = none). Thus, descriptions were available detailing which role cluster is mainly responsible for which process and where the roles provide support. The present numerical assignment is used in the further course for the identification of competencies.

0 = None 1 = Support 2 = Responsible		Role			
		Customer representative	System developer	⋮	V&V operator
Process	Architecture definition	1	2	…	0
	System requirements	2	2	…	1
	….	…	…	…	…
	Validation	1	1	…	2

Fig. 4. Role-Process-Matrix

B. *Identifying competencies for role clusters*

As competencies are the third aspect of the role cluster definition, they have to be identified and allocated to each role cluster.

a) Identifiying competencies

To identify the needed competencies, requirements for these competencies have been defined within the research project. In the following the requirements are explained:

- *Generic*: The defined competencies shall be applicable for different kinds of stakeholders as well as different kinds of companies.
- *Systems Engineering*: The defined competencies shall include all necessary skills and ways of thinking to develop a system properly.
- *Agile Development*: The defined competencies shall include the necessary skills and ways of thinking for an agile development.
- *Understandable:* The competencies shall be easy to understand by technical and non-technical stakeholders.
- *Redundancy free*: Each defined competency shall be different from other competencies.

Based on the defined requirements, the state of the art has been analyzed (cf. II B). As the INCOSE Competency Framework already includes the necessary competencies for Systems Engineering and includes a link to SE processes the defined competencies build the basis for the to-be-defined competencies. However, competencies regarding agile development have been added. Furthermore, the competencies of the INCOSE Competency Framework have been redesigned and clustered to ensure a clear understanding. The resulting set of competencies is shown in Figure 5. The competencies have been named and definitions were added. The description of the competency "System Modelling and Analyses" is, for example, "Proficiency in providing accurate data and information using models to support technical understanding and decision making".

Fig. 5. Extended INCOSE Competency Framework

The competencies have been clustered in four major groups: **Core competencies** support engineering as well as Systems Engineering. **Social / Personal competencies** cover the behavioral competencies. **Management competencies** are the competencies to perform tasks necessary to oversee and manage SE activities. **Technical competencies** include all ways of thinking, method-knowledge, and abilities to perform the technical SE processes. Agile competencies are crucial to successfully manage and execute projects and product development in an agile environment and have been considered as part of the Core, Social / Personal and Technical Competencies. The defined set of competencies includes all necessary abilities, skills, and ways of thinking to enable Systems Engineering in an agile environment. Between different roles, the necessary level of each competency may differ. This will be considered in the next step.

b) Allocation of competencies to role clusters

Competencies are needed by roles to perform their assigend tasks and processes. Therefore, the defined set of competencies has been allocated to the set of processes defined in III A. The major concern was finding out which competency was required for each process. For the mapping, a scheme with three categories was described as follows: "the competency is not needed" (= 0 = None), "the competency would be supportive" (= 1) and "the competency is mandatory" (= 3). The mapping of competencies to the processes of the ISO 15288 within the INCOSE Competency Framework [9] has been used as an Input.

0 = None 1 = Support 3 = Mandatory		Process			
		Architecture definition	System requirements	⋮	Validation
Competencies	Systems Thinking	3	3	…	1
	Communications	3	3	…	3
	….	…	…	…	…
	Operation and Support	1	1	…	0

Fig. 6. Process-Competencies-Matrix

Because not all of the existing competencies were mapped to the ISO 15288 processes and because other competencies and essential processes were missing (cf. II A), further analyses were necessary. As part of the research project, a

workshop was carried out with three industry companies to map the open competencies. Based on this analysis, a Process-Competencies-Matrix has been defined as shown in Figure 6.

Together with the allocation of roles to processes from III A, an implicit link between role clusters and competencies has been defined. The explicit allocation will be made by multiplying the matrices of "Role-Process-Matrix" and "Process-Competencies-Matrix". The result is a mapping of competencies to role clusters and even the needed competency level. The competency levels are divided into the four levels: "Knowing (= 1), Understanding (= 2), Using (= 3) and "Mastering (= 6)" according to BLOOM [16] (cf. Figure 7). If a competency is not needed, the value in the matrix is 0. For example, if a role cluster is responsible for a process (= 2) and the competency is mandatory (= 3) for the process, the result in the matrices' multiplication will be "Mastering" (= 6). The differentiation between competency levels is necessary as different role need different forms of a competency. The competencies are defined regarding different fields. For example, "Requirements Definition" is a competency that is needed for the definition of requirements. One role might need the competency to apply requirements engineering methods, while another role may only need the competency that requirements engineering is an elaborate process.

Fig. 7. Matrix multiplication leads to the desired Role-Competencies-Matrix

So, a role who needs the competency level "Knowing" shall be able to recognize and contextualize content regarding the competency (e.g. Requirements Definition). "Understanding" represents that the role is able to comprehend and interpret the content, while "Using" means that the role can independently apply the content to an equivalent situation. "Mastering" means that the role is able to adapt the required content to specific situations.

Thus, three different matrices have been defined: One for the correlation between roles and processes, another for the link between processes and competencies, and the last one represents the connection between roles and competencies

C. *Validation by industrial companies*

The templates were carried out as part of the research project with industrial companies involved. These companies are active in mechanical and plant engineering. The number of employees of the companies is between 5,000 and 21,000. All companies face the challenge of introducing and consolidating Systems Engineering. Since different companies with different focuses participate in the research project, the presented templates were validated in different fields.

The first validation field was the usage of the role cluster and the Process-Role-Matrix. The aim of the company was to identify which of their own existing roles is already performing Systems Engineering tasks and processes as well as which are to be exercised in the future. Therefore, the company-specific roles have been assigned to the role clusters. The assignment of their own roles to the role clusters was carried out by the individual roles themselves using the role cluster template. The Process-Role-Matrix provided assistance in quickly determining the necessary SE processes for each individual stakeholder. All company-specific roles were able to be assigned to the role clusters, but the assignment was not unambiguous. For example, two different roles have been mapped to the cluster "customer representative". As a result, the Process-Role Matrix had to be adapted. Those changes have been made quickly by adding the company-specific roles to the matrix and reassigning the numbers. One of the most important findings of this validation was that difficulties have emerged when the assignments have to be performed parallel to the adaptation of the Role-Process Matrix. To give guidance it is advisable to define a procedure for these tasks.

The second validation has been performed with another company that is trying to determine their as-is status of currently available Systems Engineering competencies to be able to define in future concrete qualification measures. For this purpose, the competencies defined in III B were used as the basis for a survey. Therefore, the competencies were examined regarding their suitability for the company as well as their allocation to the processes of the ISO 15288. The results of this analysis have already been considered in the presented competencies and Process-Competencies-Matrix. Furthermore, the role cluster will be used within this survey and the employees have to assign themselves to those clusters. After the survey the results are to be analyzed and qualification measures will be derived.

Nevertheless, further development potential can be identified in the present concept: First of all, a procedure of how to use those presented templates should be defined. Further, the concept is currently based fundamentally on the processes of ISO 15288. A company's deviation from these process steps is not taken into account in the present concept. In addition, the current concept does not take into account existing competency models. If a company already has certain qualification programs, it cannot be ensured that these can be linked to the concept. These three concerns have been identified so far.

IV. Conclusion and Future Work

Companies are facing new challenges to master the transformation to Advanced Systems. Introducing Systems Engineering (SE) into the company is an approach to meet these challenges in product development. Since SE is a holistic approach, all employees in the company must be involved in order to achieve a company-wide steadiness. A key question here is which competencies the individual employees in the company need for this purpose.

This paper presents a solution concept that enables industrial companies to identify stakeholder-specific Systems

Engineering competencies. The relevant SE processes are taken into account as well. The provided templates enable the companies to define SE roles, processes, and competencies. A beneficial aspect of the defined solution is that adjustments can be easily made. When the responsibilities of a process change (Role-Process-Matrix), the needed competencies within the Role-Competencies-Matrix will change immediately.

However, as the validation shows there is the need of a procedure model that supports the usage of the given templates. A preliminary idea of this procedure has already been defined and shall be presented shortly (cf. Figure 8).

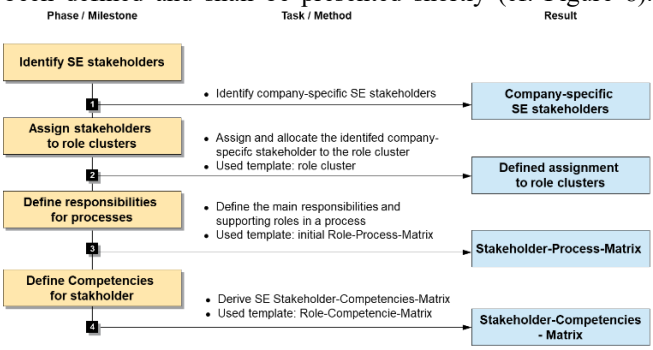

Fig. 8. Procedure for the identification of the stakeholder-specific SE-competencies

The first step is to identify all relevant stakeholders who are involved in the implementation of Systems Engineering. The focus shall be on internal stakeholders, as company external stakeholders (e.g. law, supplier or contender) do not need to perform processes within the company. The output will be a set of company-specific internal stakeholders who will contribute to SE.

In the second step, the identified set of company-specific stakeholders needs to be assigned to the provided role clusters in order to link them to the SE processes. The role clusters are a help to support this allocation. If multiple stakeholders are assigned to one cluster, the responsibilities must be redefined. This will be done within the next step.

In the third step, the responsibilities for the SE processes will be defined. This step is necessary as roles and their responsibilities for SE processes may vary strongly from company to company. The provided Role-Process-Matrix shall be used as an input. The output of this step is a Stakeholder-Process-Matrix. To differentiate the input template from the output matrix a differentiation between Role-Process-Matrix (= input) and Stakeholder-Process-Matrix (= output) has been made.

The fourth and last step will lead to competencies for each identified stakeholder. Input is the provided Process-Competencies-Matrix. As described in section III B, the matrix multiplication will lead to the Stakeholder-Competencies-Matrix. This includes the necessary competency level for each stakeholder.

In further work of the research project this procedure will be scientifically evaluated. Therefore, corresponding procedures as part of the state of the art will be analyzed and the procedure will be validated within a company. Based on this procedure, qualification programs for SE can be developed in the future. Based on the defined outcome, stakeholder-specific qualification programs can be derived. Therefore, the defined competency level of each stakeholder can be used to define concrete learning objectives and contents. These objectives and contents can at least be company-specific and could be implemented in individual qualification offerings. Through the stakeholder-specific consideration, a more effective way to address the individual needs of the roles, tailored to the necessary competencies, is to take place. This also contributes to the acceptance strategy that is being developed for the research project.

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